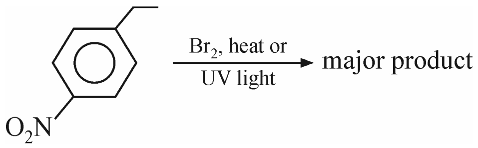
Vidyamandir Classes
Gurukul for IITJEE & Medical Preparation

illuminati-2019

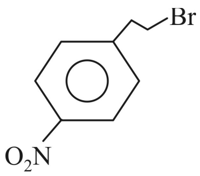
Advanced Chemistry Assignment-2B

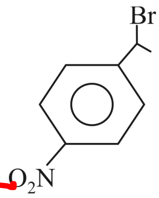
Organic Chemistry

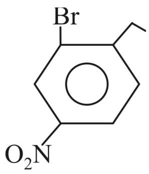
SINGLE CORRECT ANSWER TYPE

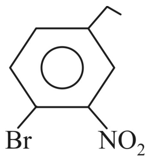
1. 

Major product is :

(A) 

(B) 

(C) 

(D) 

2. In the reactions given below :

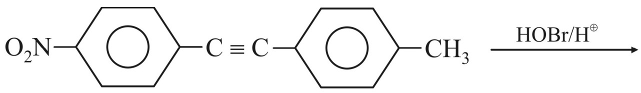
$$CH_3-CH_2Cl \xrightarrow{\text{(i) KCN (ii)LiAlH}_4} \text{Product A}$$

$$CH_3-CH_2Cl \xrightarrow{\text{(i) AgCN (ii) LiAlH}_4} \text{Product B}$$

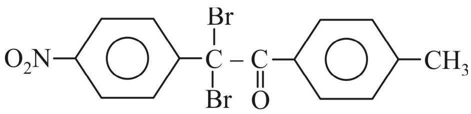
The compound A and B are:

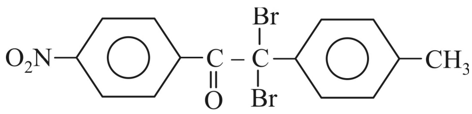
(A) chain isomers **(B)** Position isomers

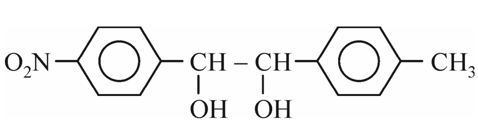
(C) Functional isomers **(D)** Metamers

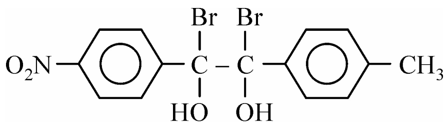
3. 

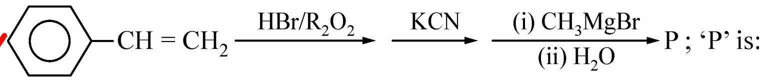
; The major product is :

(A) 

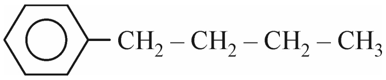
(B) 

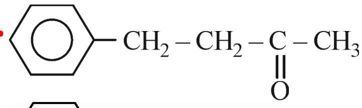
(C) 

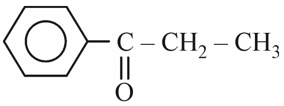
(D) 

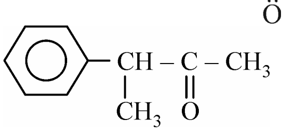
4. 

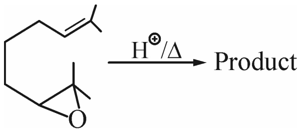
P ; 'P' is:

(A) 

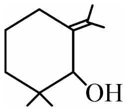
(B) 

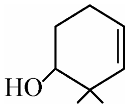
(C) 

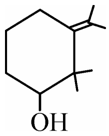
(D) 

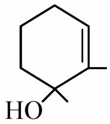
5. 

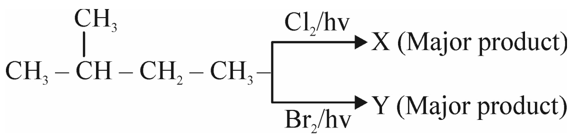
Product is :

(A) 

(B) 

(C) 

(D) 

6. 

X and Y are respectively :

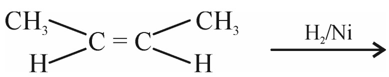
(A) $CH_3-\underset{Cl}{\overset{CH_3}{C}}-CH_2-CH_3$, $CH_3-\underset{Br}{\overset{CH_3}{C}}-CH_2-CH_3$

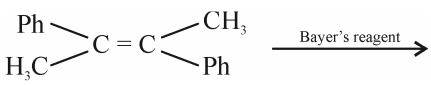
(B) $CH_3-\overset{CH_3}{CH}-\underset{Cl}{CH}-CH_3$, $CH_3-\overset{CH_3}{CH}-\underset{Br}{CH}-CH_3$

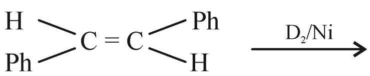
(C) $CH_3-\overset{CH_3}{CH}-\underset{Cl}{CH}-CH_3$, $CH_3-\underset{Br}{\overset{CH_3}{C}}-CH_2-CH_3$

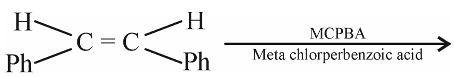
(D) $CH_3-\underset{Cl}{\overset{CH_3}{C}}-CH_2-CH_3$, $CH_3-\overset{CH_3}{CH}-\underset{Br}{CH}-CH_3$

7. Which of these reaction gives meso compound as product ?

(A) 

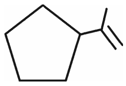
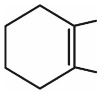
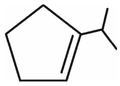
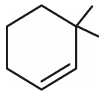
(B) 

(C) 

(D) 

8. Identify the major product (P) of the following sequence of reaction.

$$C_5H_{10} \xrightarrow{Br_2/hv} \underset{\text{(Only one isomer)}}{C_5H_9Br} \xrightarrow[\text{Ether}]{\text{Mg}} \xrightarrow[\text{(ii) } NH_4Cl]{\text{(i) Acetone}} \xrightarrow[\Delta]{H_2SO_4} P$$

(A)  **(B)**  **(C)**  **(D)** 

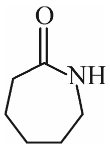
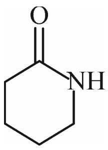
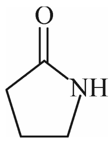
9. The pair of compounds in which both the compounds give positive test with Tollens's reagent and Fehling solution is

(A) Glucose and Sucrose **(B)** Propanal and Benzaldehyde

(C) Propanal and 2-Phenylethanal **(D)** Benzaldehyde and 1-Hydroxypropanone

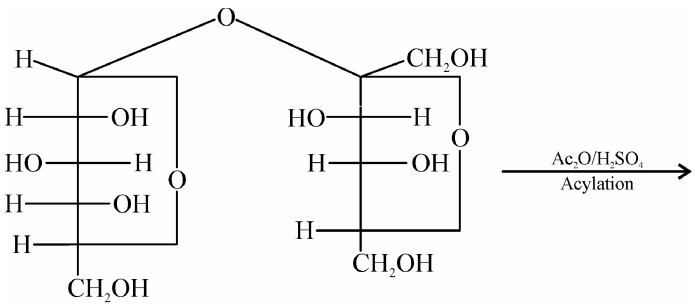
10. Which of the following is monomeric unit of the polymer Nylon-5?

$$-NH-(CH_2)_4-\underset{\underset{O}{\|}}{C}-NH-(CH_2)_4-\underset{\underset{O}{\|}}{C}-NH-(CH_2)_4-\underset{\underset{O}{\|}}{C}-$$

(A)  **~~(B)~~**  **(C)** 

(D) $NH_2(-CH_2)_4-NH_2 + HOOC-(CH_2)_3-COOH$

11. Select the correct statements about the following



(i) sucrose forms an octaacetate

(ii) sucrose is nonreducing sugar

(iii) sucrose is a polysaccharide

(A) (i), (ii) **(B)** (i), (ii), (iii) **(C)** (i), (iii) **(D)** Only (i)

12. A polypeptide contains five types of $\alpha-$amino acids. It gives following dipeptides only on hydrolysis.

$$(\text{Asp, Glu, His, Phe, Val}) \xrightarrow{\text{hydrolysis}} (\text{Val}-\text{Asp}+\text{Glu}-\text{His}+\text{Phe}-\text{Val}+\text{Asp}-\text{Glu})$$

The correct sequence of amino acids in the polypeptide is:

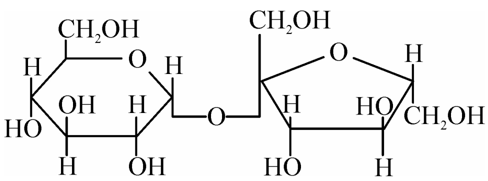
(A) Asp-Glu-His-Phe-Val

~~(B)~~ Phe-Val-Asp-Glu-His

(C) Val-Asp-Glu-His-Phe

(D) Glu-His-Phe-Val-Asp-Glu-Val-Asp

13. The following compound on hydrolysis will give :



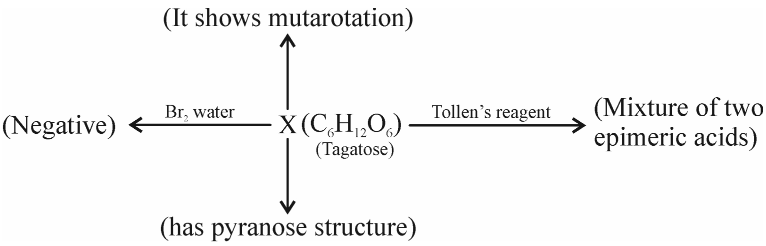
(A) A pair of anomers

(B) A pair of enantiomers

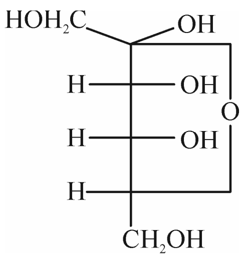
(C) A pair of epimers

~~**(D)**~~ A pair of functional isomers

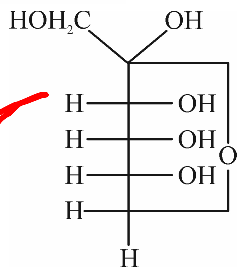
14. Observe the following road map and identify the carbohydrate 'X' (Tagatose)



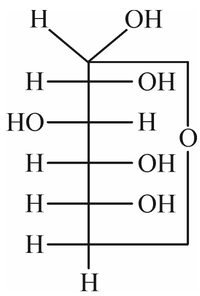
(A)



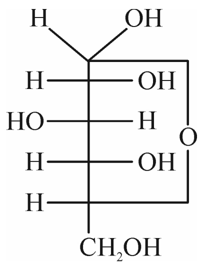
~~**(B)**~~



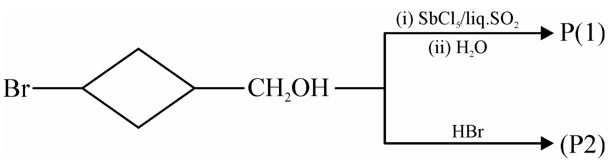
(C)



(D)



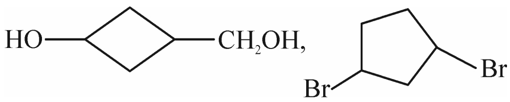
15.



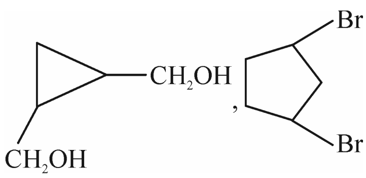
P1 and P2 are respectively :

(A)

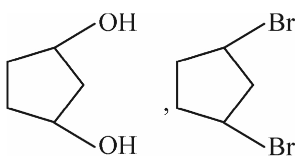
(B)



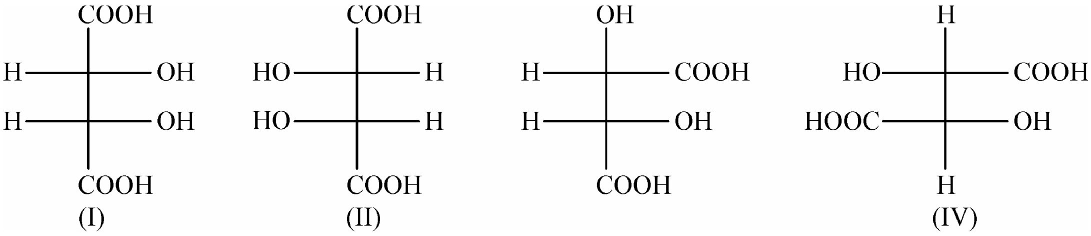
(C)



~~**(D)**~~

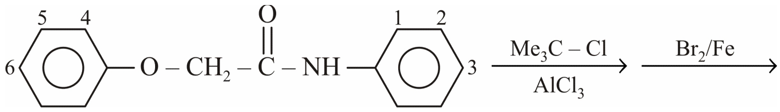


16. Which of the following structure are superimposable ?



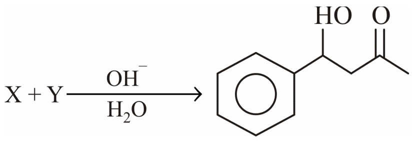
(A) I & II **(B)** II & III **(C)** III & IV **(D)** II & IV

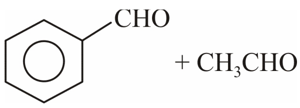
17. The main positions of electrophilic attack for the given reactions are respectively :

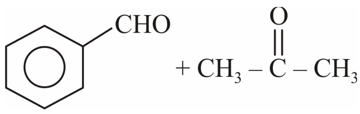


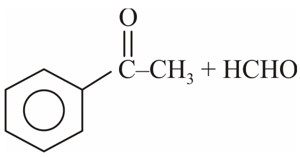
(A) 3, 1 **(B)** 6, 3 **(C)** 6, 4 **(D)** 3, 6

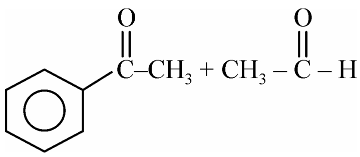
18. The reactant X & Y of following reactions are :



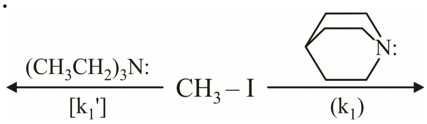
(A) 

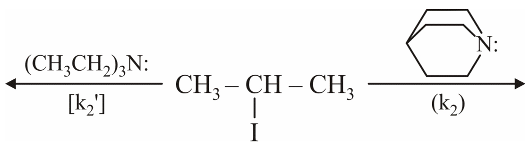
(B) 

(C) 

(D) 

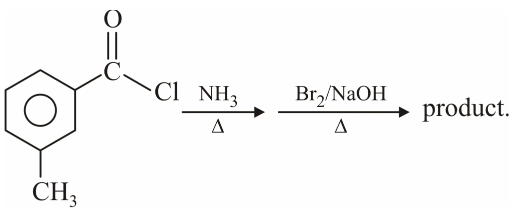
19. Observe the following reaction I and II k_1, k_1', k_2 and k_2' are rate constants. Select the correct option :

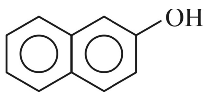
(I) 

(II) 

(A) $k_1 > k_1'$ **(B)** $k_2 < k_2'$ **(C)** $k_1 < k_2$ **(D)** $k_1' < k_2'$

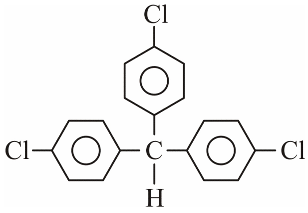
20. The product of the following reaction does not show positive test with :

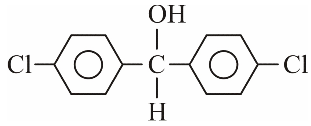


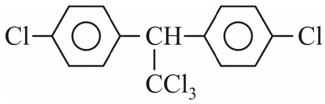
(A) $(CHCl_3 + KOH) / \Delta$ (Carbyl amine test) **(B)** HNO_2,  / OH^-, (β-naphthol test)

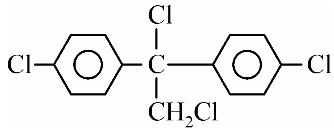
(C) $Ph - SO_2Cl$ / NaOH (Hinsberg test) **(D)** Neutral $FeCl_3$ Test

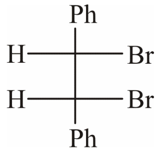
21. $CCl_3-CH=O$ reacts with chlorobenzene in presence of sulphuric acid & produces :

(A) 

(B) 

(C) 

(D) 

22. $Ph-C\equiv C-Ph \xrightarrow{x} \xrightarrow{y}$ 

The reagents x and y are respectively:

(A) $Br_2/CCl_4, H_2/Ni$

(B) HBr, HBr

(C) $Na/NH_3(\ell), Br_2/CCl_4$

(D) Br_2/CCl_4, Lindlar's hydrogenation

MULTIPLE CORRECT ANSWER TYPE

23. $R-\underset{\underset{O}{\|}}{C}-OR' \xrightarrow[2.\ H_3O^{\oplus}]{1.\ MeMgBr}$ acetone as the sole organic product. Which is/are correctly matched with R and R'.

(A) $R=-H$

(B) $R'=-\underset{\underset{CH_2}{\|}}{C}-CH_3$

(C) $R'=-\underset{\underset{CH_3}{|}}{CH}-CH_3$

(D) $R=-CH_3$

24. Which of the following would be the significant product/s of the given reaction ?

$$H-C\equiv C-\underset{\underset{OH}{|}}{CH}-CH=CH_2 \xrightarrow{H^{\oplus}(\text{traces})}$$

(A) $CH_2=CH-\underset{\underset{O}{\|}}{C}-CH=CH_2$

(B) $CH_2=CH-CH=CH-CHO$

(C) $H-C\equiv C-\underset{\underset{O}{\|}}{C}-CH_2-CH_3$

(D) $\underset{\underset{OH}{|}}{CH_2}-CH=CH-C\equiv CH$

25. Select the chain propagation steps in the free radical chlorination of methane ?

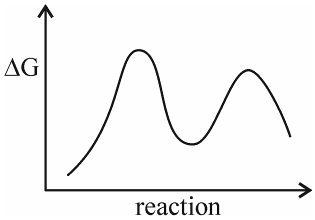
(A) $Cl_2 \longrightarrow 2\dot{C}l$

(B) $\dot{C}H_3+Cl_2 \longrightarrow CH_3Cl+\dot{C}l$

(C) $\dot{C}H_3+\dot{C}l \longrightarrow CH_3-Cl$

(D) $R\dot{O}+CH_4 \longrightarrow ROH+\dot{C}H_3$

26. Following graph between ΔG and reaction progress represented by which of the following reaction ?



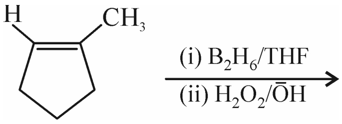
(A) S_N1 reaction

(B) E_1 reaction

(C) Aromatic electrophilic substitution

(D) electrophilic addition reaction

27.



True statement about above reaction

(A) Reagent involve stereospecific SYN addition of H and OH species

(B) Product obtained is trans isomer

(C) BH_3 act as electrophile

(D) Two stereoisomers are obtained as product

28. Select the correct statements out of the following:

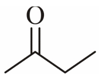
(A) $\beta-D(+)$ glucopyranose is more stable than $\alpha-D(+)$ glucopyranose

(B) sucrose is laevorotatory sugar

(C) Although fructose is a ketone but it gives positive Tollen's test

(D) Starch contains $\alpha-$glycosidic linkage whereas cellulose contains $\beta-$glycosidic linkage

29. 2-Phenylbutan-2-ol can be prepared by :

(A) PhMgBr +  $\xrightarrow{\text{ether}} \xrightarrow{H^{\oplus}}$

(B) $CH_3MgBr + Ph - \overset{\overset{\large O}{||}}{C} - C_2H_5 \xrightarrow{\text{ether}} \xrightarrow{H^{\oplus}}$

(C) $C_2H_5MgBr + Ph - \overset{\overset{\large O}{||}}{C} - CH_3 \xrightarrow{\text{ether}} \xrightarrow{H^{\oplus}}$

(D) $CH_3CH_2CH_2MgBr + PhCHO \xrightarrow{\text{ether}} \xrightarrow{H^{\oplus}}$

ASSERSION & REASON TYPE

(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1

(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1

(C) Statement-1 is True, Statement -2 is False

(D) Statement-1 is False, Statement-2 is True

30. **Statement-1:** 2-Bromobutane on heating with alcoholic KOH gives 3 isomeric alkenes which on treatment with Br_2/CCl_4 give 5 products.

Statement-2: Stereochemical nature of elimination by alcoholic KOH and addition of Br_2 is ANTI and SYN respectively.

31. **Statement-1:** The rate of catalytic hydrogenation of ethyne at raney nickel (finally divided Ni) is faster than that of ethene.

Statement-2: Catalytic hydrogenation of ethyne to ethane is less exothermic than that of ethene to ethane.

32. **Statement-1:** Hydrolysis of sucrose brings a change in sign of rotation towards plane polarized light.

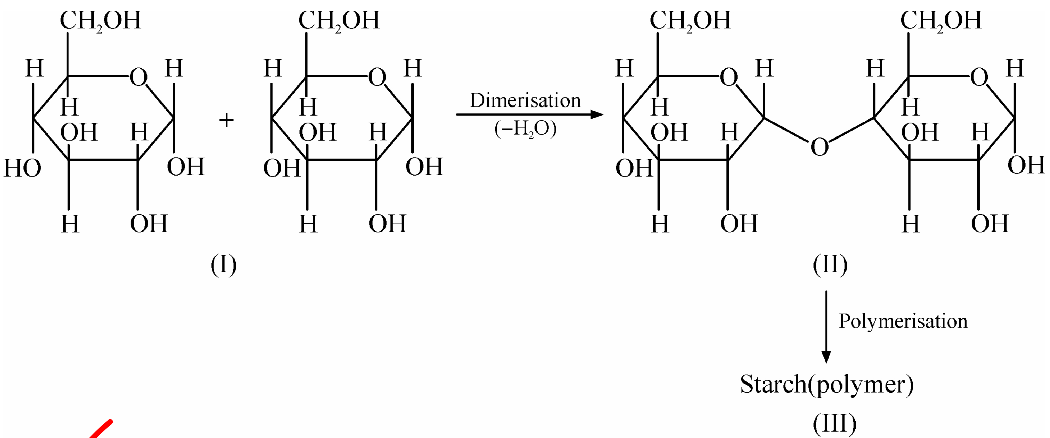
Statement-2: Fructose has specific rotation –92.4° and glucose has $+52.5^{o}$.

33. **Statement-1:** Nitration of benzene and hexadeuterobenzene occurs almost at the same rate.

Statement-2: Cleavage of C-H bond takes place faster than C – D bond.

COMPREHENSION TYPE

Comprehension-1



(I) (II) (III)

34. What is true about compound (I) :

(A) It has an acetal structure

(B) It has tertiary hydroxy group

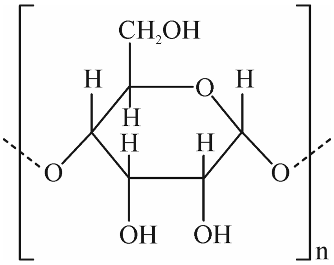
(C) It has a hemiacetal structure

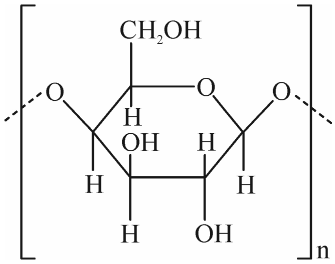
(D) It's degree of unsaturation is two

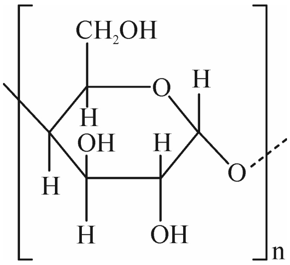
35. Compound (II) is/has :

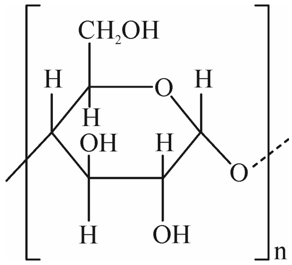
(A) A polysaccharide
(B) Oligosaccharide
(C) Monosaccharide
(D) Hydrogen deficiency index three

36. Assume that polymerization of (I) takes place in the manner similar to its dimerization, then the structure of polymer (III) can be correctly represented as

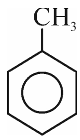
(A) 

(B) 

(C) 

(D) 

Comprehension-2



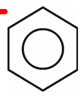
$\xrightarrow{KMnO_4} P \xrightarrow[\Delta]{NaOH/CaO} Q \xrightarrow{Na/NH_3} R \xrightarrow{O_3/Zn} S \xrightarrow{Red\ P/HI} T \xrightarrow{Br_2/hv} U$

(Major product) $\xrightarrow{Et_2CuLi} V$

37. When 'S' reacted with PhMgBr the product formed is :

(A) $Ph-\underset{OH}{\underset{|}{CH}}-CH_2-\underset{OH}{\underset{|}{CH}}-Ph$

(B) $Ph-\underset{OH}{\underset{|}{CH}}-CH_2-CHO$

(C) 

(D) $Ph-\underset{OH}{\underset{|}{CH}}-Ph$

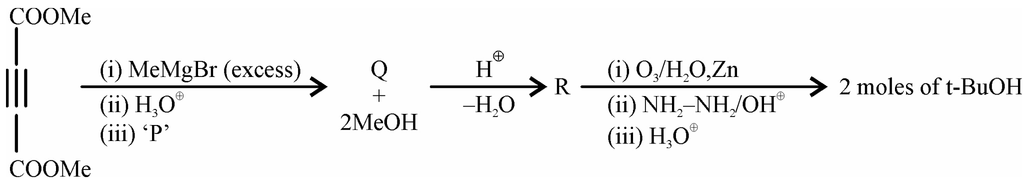
38. The total number of monochloro products of compound 'V' and the fractional distillation of product mixture respectively are :

(A) 4 , 4 **(B)** 6 , 5 **(C)** 6 , 4 **(D)** 4, 2

39. Compounds which liberate H_2 gas with Na metal are :

(A) P, S **(B)** P **(C)** P, R, S **(D)** S, V

Comprehension-3

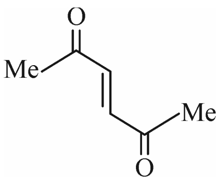


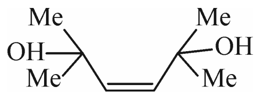
40. Reagent 'P' is :

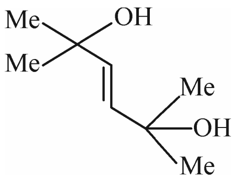
(A) $Na / NH_3(\ell)$ **(B)** H_2, Pd – $BaSO_4$ (quinoline)

(C) H_2 / Ni **(D)** $NaBH_4$

41. The structure of the product Q is :

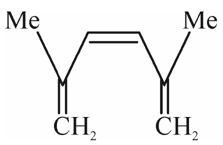
(A) 

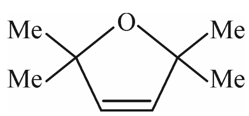
(B) 

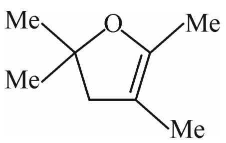
(C) 

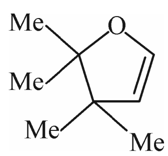
(D) $Me-\underset{OH}{\overset{Me}{C}}-C\equiv C-\underset{OH}{\overset{Me}{C}}-Me$

42. The structure of R is :

(A) 

(B) 

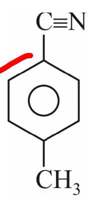
(C) 

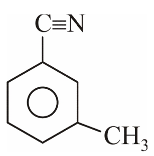
(D) 

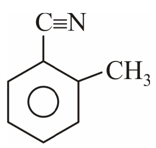
Comprehension-4

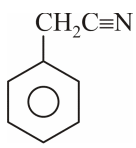
Compound (X) (C_8H_7N) on treatment with cold $SnCl_2$ / dil.HCl (aqueous), produces $Y(C_8H_8O)$. (Y) on heating with sodium acetate in acetic anhydride followed by acidification forms (Z) $(C_{10}H_{10}O_2)$. All these compounds (X,Y,Z) on oxidation under severe conditions form a diacid which gives only one mononitro isomer in nitrating mixture.

43. What is structure of (X) :

(A) 

(B) 

(C) 

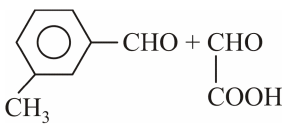
(D) 

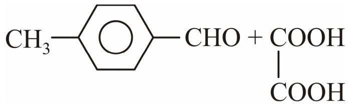
44. What is not true about (Y) :

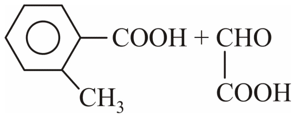
(A) It cannot gives Fehling solution test **(B)** It gives Iodoform test positive

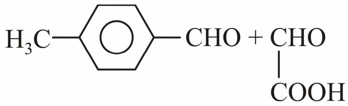
(C) It gives Tollen's test positive **(D)** it gives 2, 4-DNP test positive

45. The reductive ozonolysis products of (Z) are :

(A) 

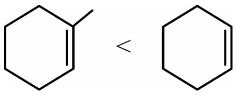
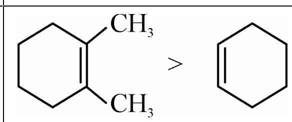
(B) 

(C) 

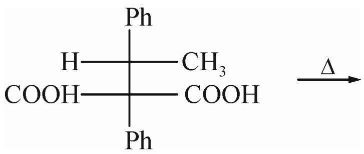
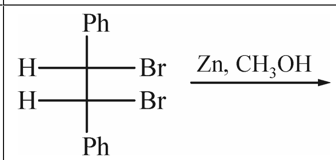
(D) 

MATRIX MATCHING TYPE

46.

	Column-I qsr		**Column-II**
(A)	$CH_3-C\equiv C-CH_3 > CH_3-CH=CH-CH_3$	**(p)**	Rate of HBr addition
(B)	 qr	**(q)**	Rate of H_2 addition
(C)	$CH_3-CH=CH_2 > CH_3-C\equiv CH$ ps	**(r)**	ΔH of hydrogenation
(D)	 ps	**(s)**	ΔH of combustion

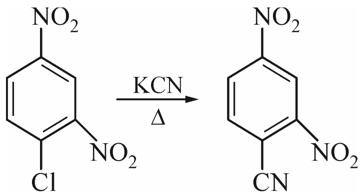
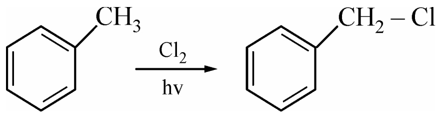
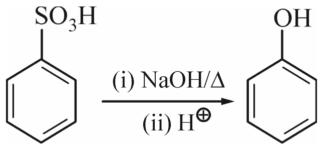
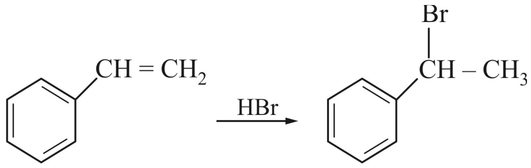
47. Match the reactions given in column I with their properties listed in column II.

	Column-I		**Column-II**
(A)	 sr	**(p)**	Anti elimination
(B)	$Ph-C\equiv C-Ph \xrightarrow[\text{poisoned}]{H_2.Pd-BaSO_4}$ q	**(q)**	Stereoselective
(C)	 p	**(r)**	Optical diastereomers
(D)	$Ph-\underset{\underset{O}{\Vert}}{C}-\underset{\underset{CH_3}{\vert}}{CD}-\underset{\underset{O}{\Vert}}{C}-OH \xrightarrow{\Delta}$ st	**(s)**	CO_2 will evolve
		(t)	Racemic mixture

48.

	Column-I		**Column-II**
(A)	$CH_3-C\equiv CH$	**(p)**	Undergoes oxidation
(B)	$CH_3-\overset{\overset{\displaystyle O}{||}}{C}-H$	**(q)**	Undergoes reduction
(C)	HCOOH	**(r)**	Gives ppt. with $AgNO_3 / NH_4OH$
(D)	$CH_3-CH=CH_2$	**(s)**	Can be reduced with help of H_2 / Pt (catalytic hydrogenation)
		(t)	more acidic than water

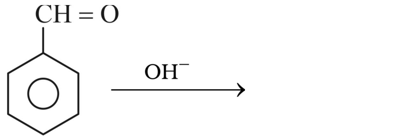
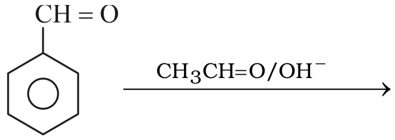
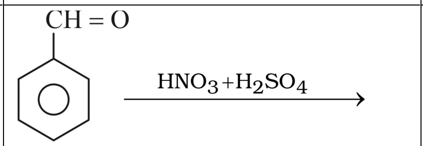
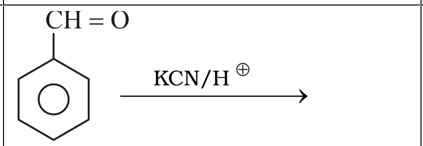
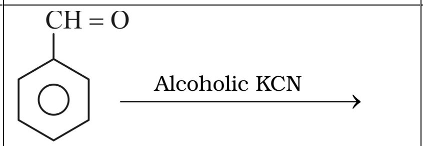
49.

	Column-I		**Column-II**
(A)		**(p)**	Substitution reaction
(B)		**(q)**	Free radical reaction
(C)		**(r)**	Nucleophile is the attacking species
(D)		**(s)**	Electrophile is the attacking species
		(t)	Addition reaction

50. Match list-I (reaction) with List-II (products)

	List-I		**List-II**
(A)	Phenol + NaOH + CH_3I	**(p)**	Phenolphthalein
(B)	Phenol + $NaOH + CHCl_3$	**(q)**	Salicylic acid
(C)	Phenol + Phthalic anhydride + conc. H_2SO_4	**(r)**	Anisole
(D)	Phenol + CO_2 + NaOH	**(s)**	Salicylaldehyde

51. Match list-I (reaction) with List-II (products) and then select the correct answer from the codes given below the lists :

	List-I		**List-II**
(A)		**(p)**	Aromatic ring substitution
(B)		**(q)**	Nucleophilic addition
(C)		**(r)**	Condensation reaction
(D)		**(s)**	formation of more than one organic products
(E)			

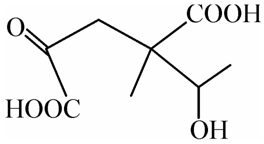
SUBJECTIVE TYPE

52. The sum of structural and diastereomers of a hydroxy compound C_4H_8O, formed by the substation reaction of 3-chlorobut-1-ene with $AgNO_3$ solution.

53. How many alkenes (be sure to consider the stereochemistry of the alkene) can be hydrogenated to form methyl cyclohexane

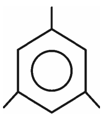
6

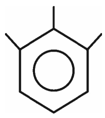
54. An optically active compound (M) on ozonlysis gives following structure in absence of zinc.

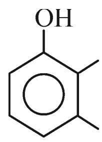


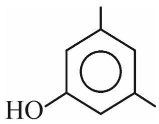
If (M) does not responds to $NaHCO_3$ test, what product is obtained when (M) is heated in presence of conc. H_2SO_4 as catalyst. Fill the number against product in key :

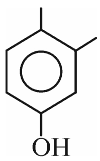
5

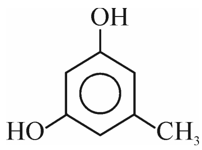
(1) 

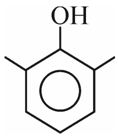
(2) 

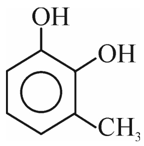
(3) 

(4) 

(5) 

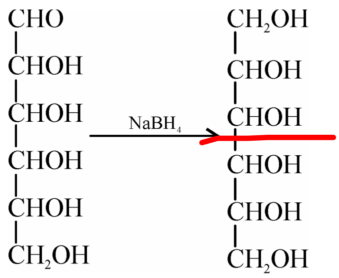
(6) 

(7) 

(8) 

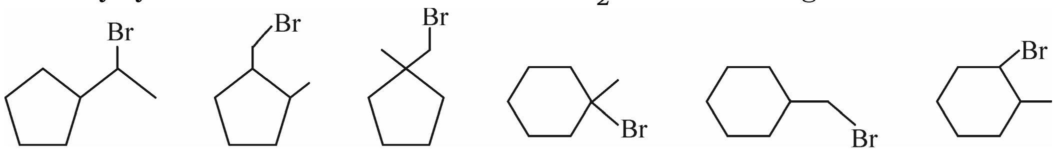
55. Lithium-di-(3-pentyl)cuperate on reaction with an alkylbromide produces (X) C_7H_{16}. The mixture of all monochloro isomers formed from (X) are subjected to fractional distillation. Find the number of fractions obtained.

56. Observe the following reaction and find out that how many number of reactant stereoisomers can be reduced to optically inactive meso products.



4

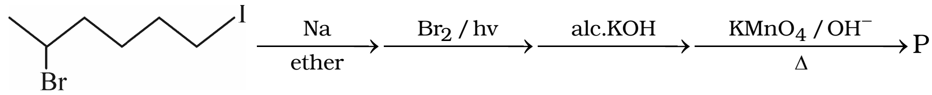
57. Among the 6, how many cyclic isomers of molecular formula $C_7H_{13}Br$ can form 1-methylcyclohexane-1-ol on reaction with H_2O /acetone / Ag^+



58. An undergraduate student heated 2-Cyclopentylethyl ethanoate at 300°C to obtain an unsaturated hydrocarbon "K" which was then treated with mercuric acetate followed by water then sodium borohydride to give "L" which is dehydrated by concentrated H_2SO_4 at 180° C to give "M" as major product. Find molecular mass of compound "M"

59. Ozonolysis of next higher homolog of simplest alkene in presence of zinc yields (A) and (B). (A) gives positive iodoform test while (B) does not. (A) when treated with excess of (B) in NaOH solution gives (C) and (D) as final product after acidification. If "C" gives positive $NaHCO_3$ test then calculate molecular mass of (D). Report your answer as molecular mass divided by 4.

60. Observe the following sequence of reactions and report your answer as molecular weight/10



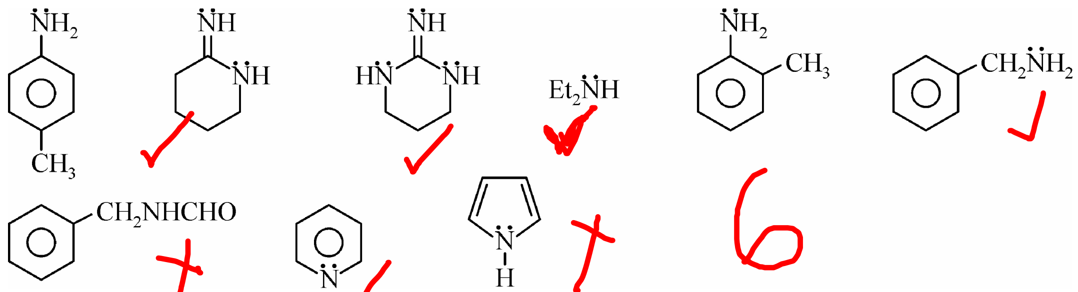
13

61. $CH_3-CH_2-CH_2-\underset{\underset{OH}{|}}{CH}-CH_3 \xrightarrow[-H_2O]{H^+} [F] \xrightarrow{Br_2/CCl_4} C_5H_{10}Br_2\ (G)$

6

How many total products of (G) are formed ?

62. How many of the following are stronger bases than aniline ?



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63. 3-Amino-2, 3-dimethyl butan-2-ol when treated with $NaNO_2$ / HCl gives product (X). (X) on treatment with $LiAlH_4$ forms (Y). (Y) on heating with conc. H_2SO_4 forms (Z). calculate the molecular weight of (Z).

Answer Key
Advanced Chemistry Assignment-2B
Organic Chemistry

1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**
B	C	B	B	C	C	D	B	C	B	A	B
13	**14**	**15**	**16**	**17**	**18**	**19**	**20**	**21**	**22**	**23**	**24**
D	B	C	A	C	B	A	D	C	C	BD	BD
25	**26**	**27**	**28**	**29**	**30**	**31**	**32**	**33**	**34**	**35**	**36**
BD	ABCD	ABCD	ACD	ABC	C	C	A	B	C	B	D
37	**38**	**39**	**40**	**41**	**42**	**43**	**44**	**45**	**46**		
C	C	A	B	B	B	A	B	D	A-qr ; B-qr ; C-ps ; D-ps		
47			**48**					**49**			
A-rs ; B-q ; C-pq ; D-st			A-pqrs ; B-pqrs ; C-pqrt ; D-pqs					A-pr ; B-pq ; C-pr ; D-st			
50			**51**				**52**	**53**	**54**	**55**	**56**
A-r ; B-s ; C-p ; D-q			A-qs ; B-qrs ; C-p ; D-qs ; E-qrs				4	6	5	3	4
57	**58**	**59**	**60**	**61**	**62**	**63**					
6	96	34	13	6	6	84					